Evaluation of HMA with RAS and Fibers

Performance Based Evaluation of Polymer Fiber Reinforced Hot Mix Asphalt with Reclaimed Asphalt Shingles

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Outline

- Why This Study Matters
- Objective
- Experimental Program
- Key Findings
- Bottom-Line Conclusions
- Research Gaps / Recommended Future Work

Introduction

The transition toward sustainable and durable pavement design.

Highway Pavement Demands

Highway agencies face challenges maintaining networks while achieving sustainability.

California’s Senate Bill 1 (SB1) pushes for materials that last longer and reduce environmental impacts.

Traditional volumetric design is insufficient for today’s heavy traffic.

RAS Usage

- Reclaimed asphalt pavement (RAP) and shingles (RAS) support sustainability by reducing virgin material.
- The U.S. generates over 13 million tons of waste shingles annually, a massive recycling opportunity.
- RAS contains highly aged asphalt binder, allowing it to replace virgin binder and aggregate. (19-36% binder content)
- However, this highly oxidized binder creates critical mixture design challenges.

Polymer Fiber Reinforcement

Polymer fibers can be introduced to the asphalt matrix to prevent premature pavement failure while using RAS.

High-temperature-stable synthetic fibers, like para-aramid, reinforce the mastic.

They bridge micro-cracks before they propagate, extending pavement life.

Benefits of Polymer Fibers

Crack Bridging

Fibers spread throughout the mastic, intercepting micro-cracks and delaying fracture propagation under repeated loads.

Rut Resistance

Three-dimensional reinforcement networks lock aggregates in place, improving stability and high-temperature deformation resistance.

Adaptive Design

Fibers restore flexibility to stiff RAS mixtures, achieving the critical balance required in modern performance-based designs.

Problem Statement & Research Gap

PROBLEM

Core Problem: High RAS content increases mixture stiffness and brittleness, leading to poor cracking resistance and difficulty in achieving balanced asphalt performance.

Identified Research Gaps:

- **Limited data on combined RAS + polymer fiber effects in dense-graded HMA under BMD testing protocols**
- **Unclear whether fiber reinforcement can simultaneously restore cracking resistance without compromising rutting**
- **No published study specifically evaluates two para-aramid fiber types at multiple dosages with 25% RAP + varying RAS**

Key Takeaway: A direct performance comparison of fiber type, dosage, and RAS levels is critically needed.

Research Objectives

Validating the efficacy of fibers in high-recycled mixes.

Primary Study Goals

- **Evaluate Rutting & Moisture:** Assess the effect of polymer fiber type and dosage on the rutting and moisture resistance of asphalt mixtures containing 25% RAP and varying RAS (3% and 5%) using HWT.
- **Assess Cracking Resistance:** Determine the influence of fiber reinforcement on intermediate-temperature cracking resistance of RAS-modified mixtures via the IDEAL Cracking Test.
- **Restore Balanced Performance:** Determine if polymer fiber reinforcement can restore balanced performance in high-RAS mixtures within a performance-based mix design framework.

Materials and Methodology

Properties, components, and laboratory testing procedures.

Aggregate Properties (3% & 5% RAS)

- **Source: Granite Construction**
- **Used in active Caltrans projects in the Bay Area**

Caltrans 2024 Standard Specifications

PROPERTY	3% RAS	5% RAS	CALTRANS REQUIREMENT	STATUS
Crushed Coarse (1 fractured face), %	98	99	≥ 95	✓ Pass
Crushed Fine (1 fractured face), %	96	99	≥ 40	✓ Pass
LA Abrasion (500 rev), % loss	17	20	≤ 40	✓ Pass
Sand Equivalent	80	76	≥ 47	✓ Pass
Bulk Specific Gravity (Blend)	2.666	2.66	—	—

Asphalt Binder Properties

PG 58-22

Performance Grade Binder

Key Characteristics

- Specific Gravity: 1.0264 (AASHTO T 228)
- High-Temp Rheology: G*/sinδ = 3.92 kPa on RTFO residue, exceeding the 2.20 kPa minimum.
- Mixing Temperature: 289°F to 299°F.
- Antistrip Additive: Evoflex X1 included at 0.5% by weight of the asphalt binder.

Polymer Fiber Types A and B

- Supplier: Surface Tech
- Both fibers were evaluated at dosages of 0.05%, 0.10%, and 0.15%.

Fiber Type A

Para-aramid synthetic fiber (minimum 63% by weight) combined with a wax treatment (maximum 37% by weight). High decomposition temperature (>425°C) and robust tensile strength.

Fiber Type B

Para-aramid synthetic fiber (minimum 75% by weight) treated with a liquid emulsion binder (maximum 25% by weight). Shares identical tensile strength and thermal resilience specifications.

Mix Preparation & Testing

- Constant RAP: Maintained at 25% by weight.
- Variable RAS: Incorporated at 3% or 5%.
- Optimal Binder Content: Held constant at 5.2% by total mix weight for all mixtures.
- Mixing: Aggregates, RAP, and RAS heated, then mechanically mixed with binder and fibers until uniformly coated.

Gyratory Compaction Process

Specimen Fabrication

Specimens were prepared using a Superpave Gyratory Compactor.

Specimens were compacted to a target air void level of 7.0 ± 0.5%.

Target heights were 60 mm for HWT and 62 mm for IDEAL-CT.

Volumetrics & Specific Gravity

7.0%

Target Air Voids (±0.5%)

Theoretical Maximum Specific Gravity

The TMSG was determined using AASHTO T 209 and calculated with the following equation:

$$TMSG = \frac{A}{A + B - C}$$

Experimental Testing Setup

Performance evaluation of rutting, moisture, and cracking resistance.

Hamburg Wheel Tracking Test

- **Standard: Conducted per AASHTO T 324.**
- **Conditioning: Submerged in water bath at 50°C.**
- **Loading: Repeated steel wheel passes applied to track vertical deformation.**
- **Metrics: Rut depth at 15,000 passes and Stripping Inflection Point (SIP) for moisture damage.**

IDEAL Cracking Test (CTindex)

Intermediate Temp Cracking

Cracking resistance was evaluated using the IDEAL Cracking Test (ASTM-D8225).

Specimens were conditioned at intermediate temperature (22°C).

$$PG\ IT = \frac{PG\ HT + PG\ LT}{2} + 4$$

Axial loading applied at 50 mm/min until failure.

Load and displacement data determined the Cracking Tolerance Index.

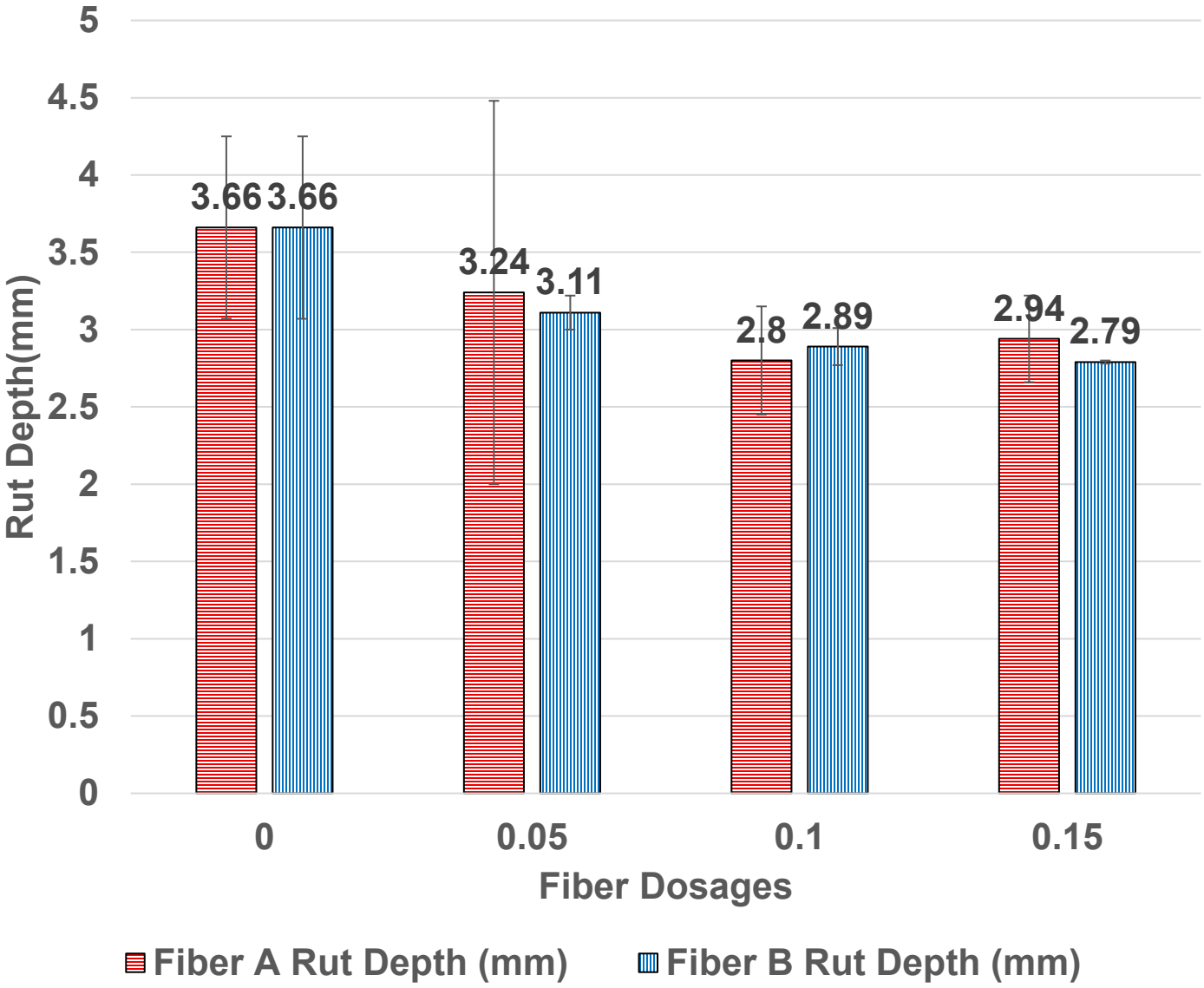
HWT Results - Rutting

Evaluation of high-temperature deformation resistance.

Rut Depth

(3% RAS)

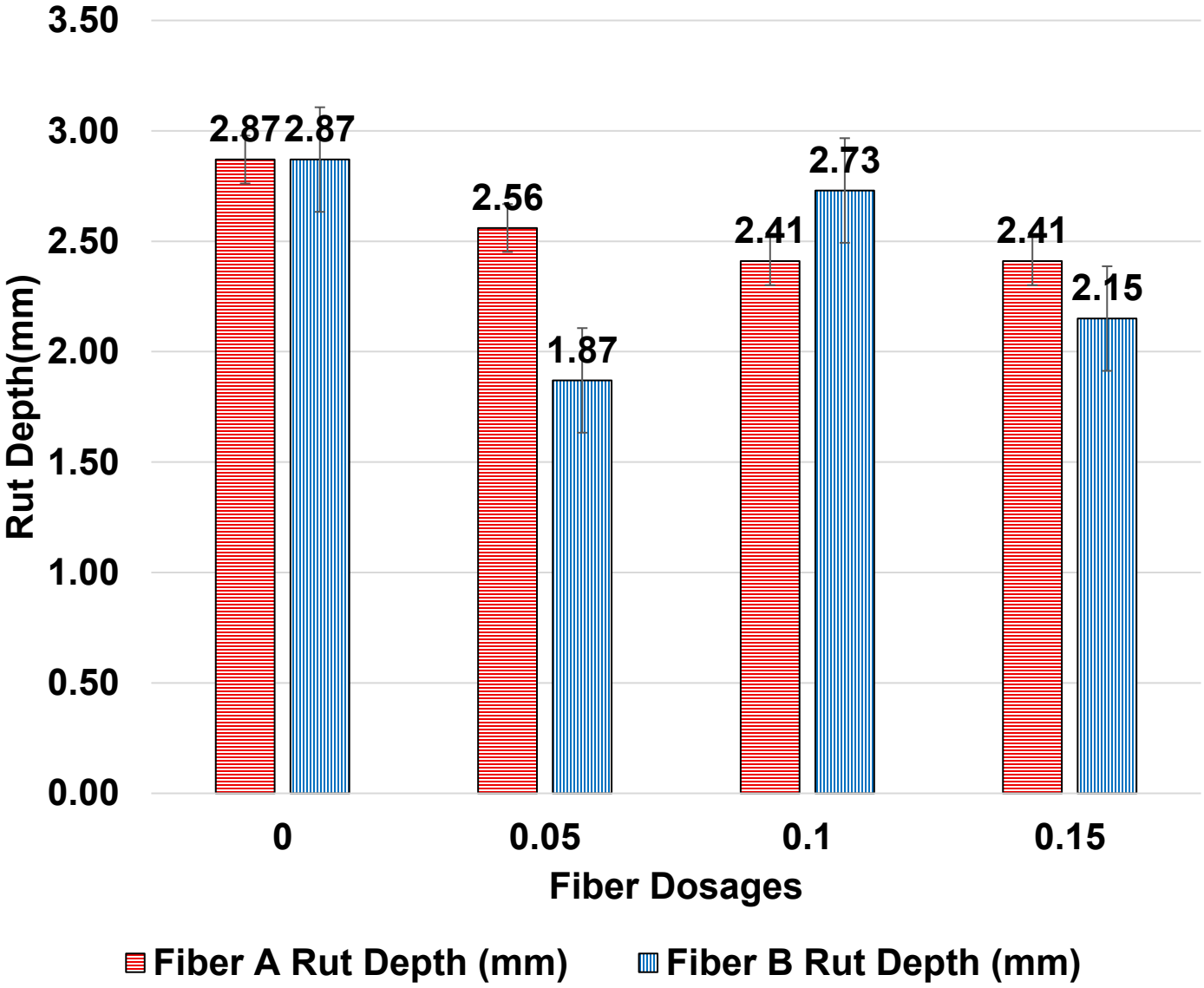
- *At 15,000 passes, both Fiber A and Fiber B improved rutting resistance over the control, with optimal reductions occurring around 0.10% to 0.15% dosage.*



Rut Depth

(5% RAS)

- *Higher RAS content lowered baseline rutting. Fiber B demonstrated pronounced improvement, notably reducing rutting to 1.87 mm at a 0.05% dosage.*



Moisture Resistance: Stripping Inflection Point Results

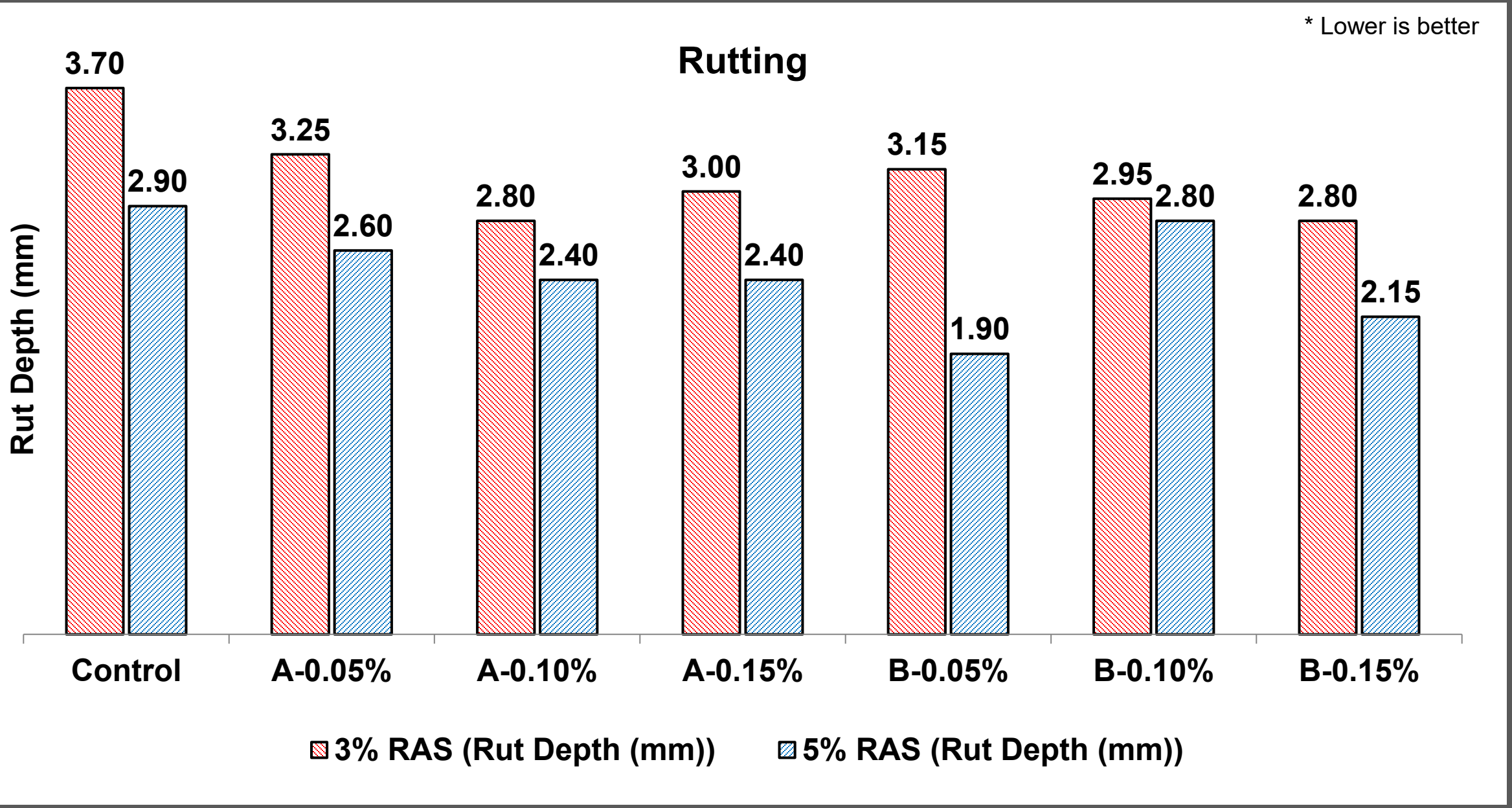
RESULTS

Stripping Inflection Point (SIP) for All Mix Combinations

✓ **All 14 mix combinations exhibited SIP > 10,000-wheel passes**

✓ **Met the AASHTO T324 moisture resistance threshold**

- No stripping was observed before 10,000 passes in any fiber type, dosage, or RAS level combination
- Several fiber-modified mixes showed no stripping point at all within the full 15,000-pass test range
- Evoflex X1 recycling agent + antistrip additive at 0.5% likely contributed to strong moisture resistance
- Higher RAS content (5%) did not introduce additional moisture susceptibility: contradicting concerns in some literature
- Fiber incorporation at 0.05–0.15% did not compromise moisture resistance at either RAS level


* Lower is better
Rutting
Rut Depth (mm)
3.70
2.90
3.25
2.60
2.80
2.40
3.00
2.40
3.15
1.90
2.95
2.80
2.80
2.15
Control
A-0.05%
A-0.10%
A-0.15%
B-0.05%
B-0.10%
B-0.15%
3% RAS (Rut Depth (mm))
5% RAS (Rut Depth (mm))

ANOVA Results - Rutting

RAS %	Fiber Type	Dosage	Mean Rut (mm)	Significance ($p < 0.05$) vs Control
5%	Control	0%	2.87	N/A
5%	Type A	0.10%	2.41	✓ Yes (p=0.0267)
5%	Type A	0.15%	2.41	✓ Yes (p=0.0267)
5%	Type B	0.05%	1.87	✓ Yes (p=0.011)
5%	Type B	0.15%	2.15	✓ Yes (p=0.0349)

* Note: 3% RAS reductions were observed but were not statistically significant at the 95% confidence level.

Hamburg Wheel Tracker Summary

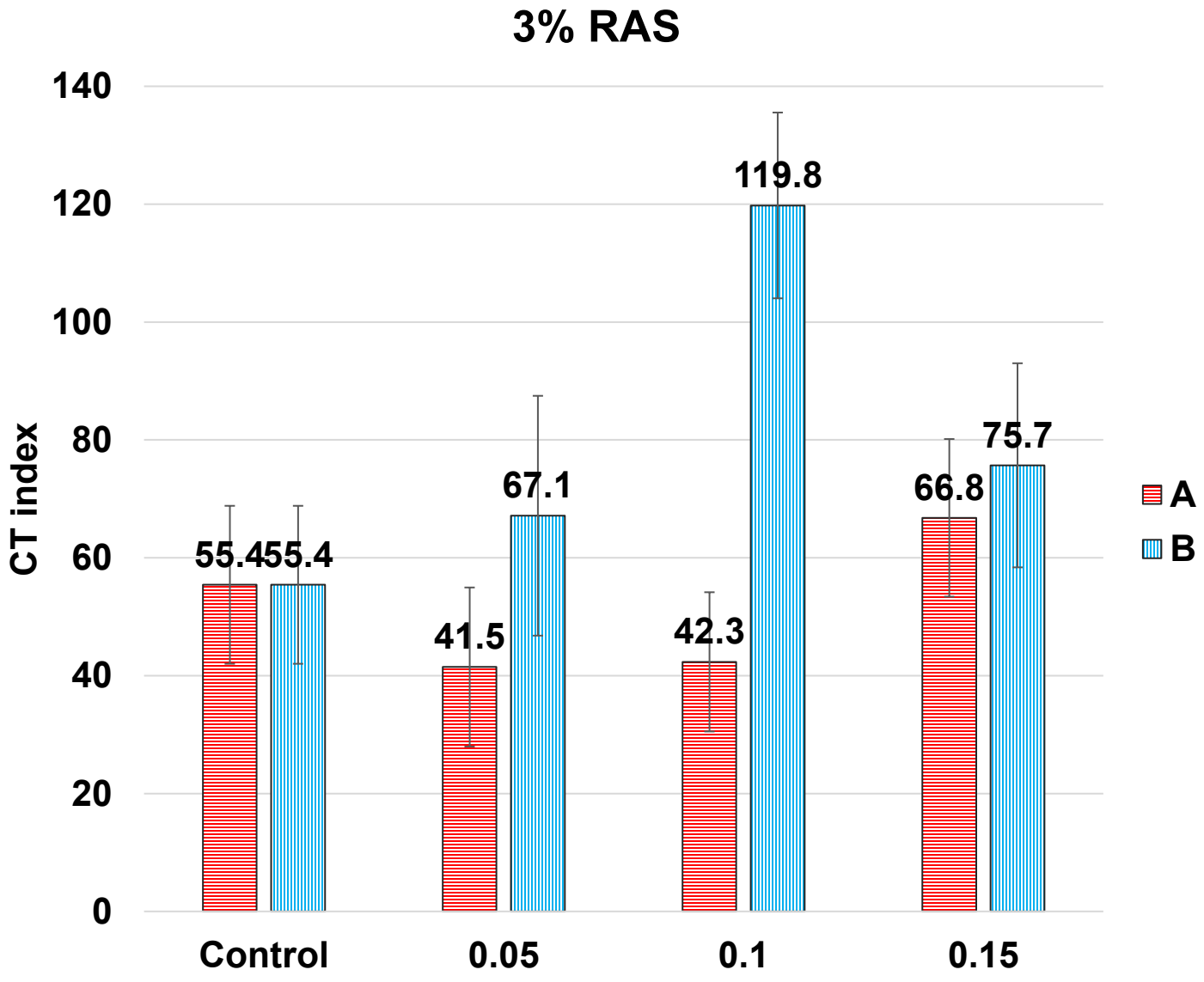
- RAS Effect: Increasing RAS content from 3% to 5% naturally reduced baseline rut depth from 3.66 mm to 2.87 mm.
- Fiber Reinforcement: Fibers further enhanced rut resistance across both RAS levels without negatively impacting volumetrics.
- Type B Superiority: Fiber Type B generally exhibited better resistance to rutting than Type A, especially in the 5% RAS mixtures.
- Moisture Durability: Stripping resistance remained universally satisfactory (>10,000 passes) regardless of fiber dosing or RAS increase.

IDEAL-CT Results - Cracking

Evaluating intermediate-temperature cracking resistance and fracture tolerance using CTindex.

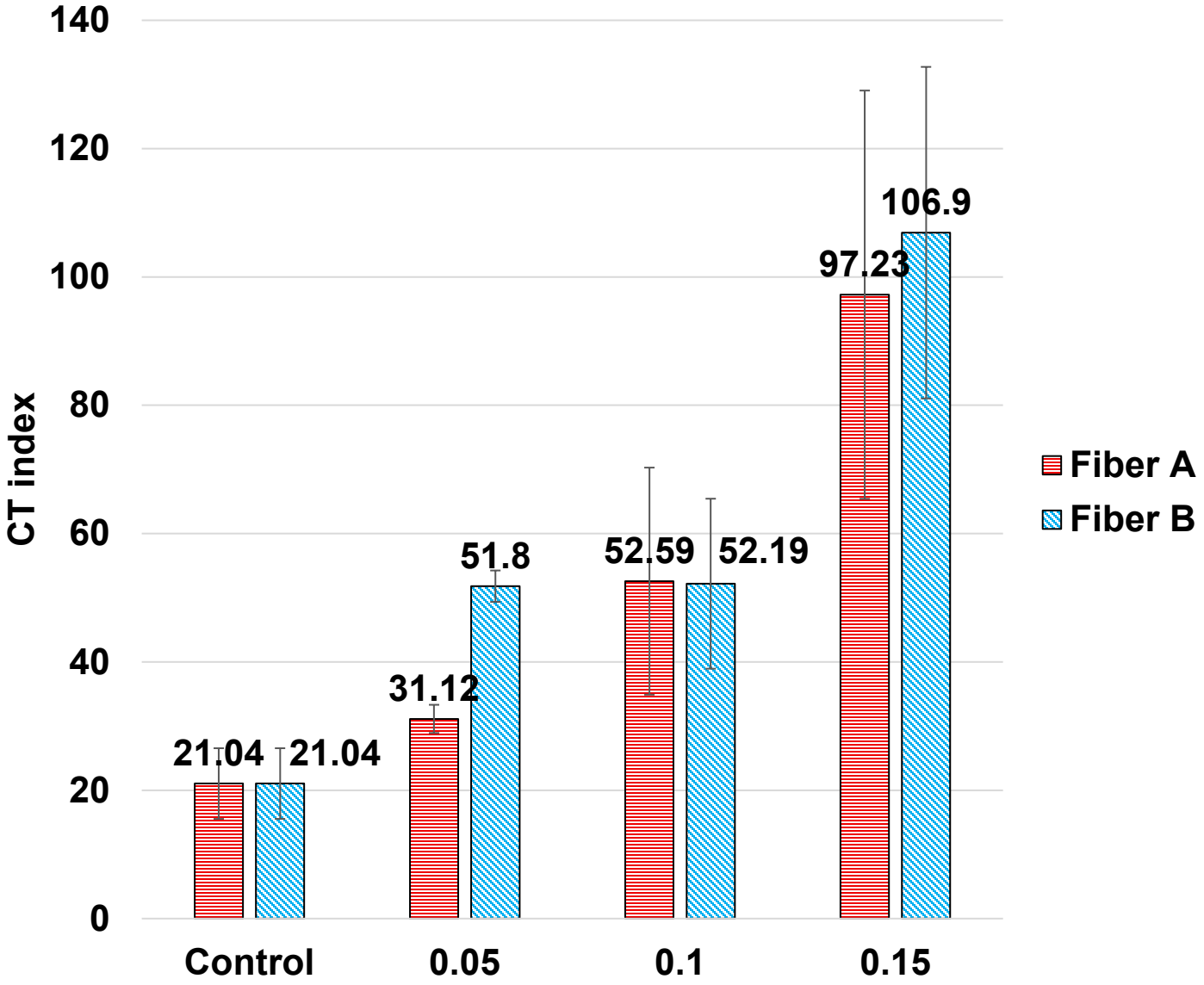
CTindex Results (3% RAS)

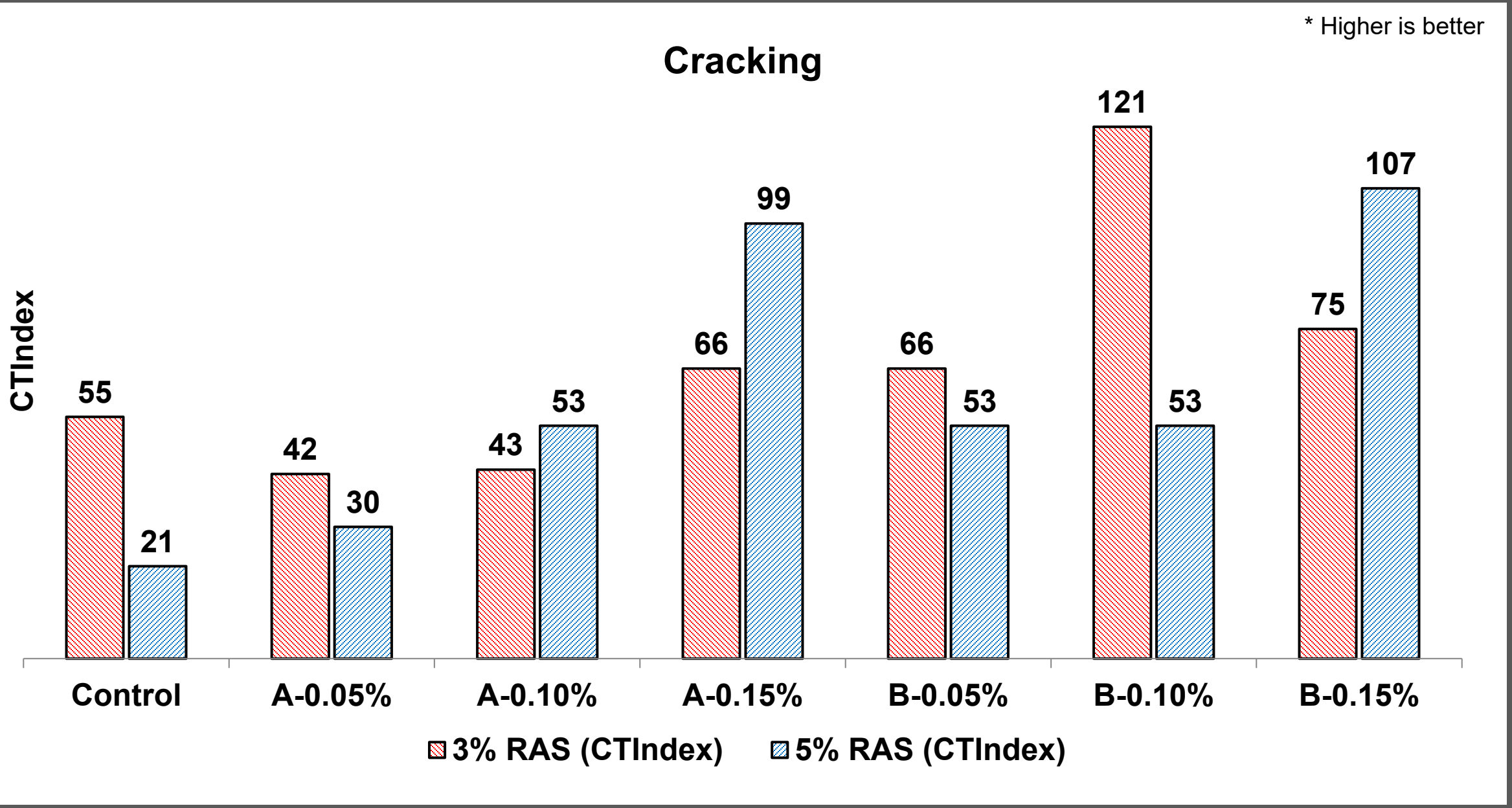
- *Fiber B at 0.10% produced a substantial enhancement in cracking tolerance, more than doubling the baseline CTindex value of the control mixture.*



CTindex Results (5% RAS)

- *The 5% RAS control was extremely brittle (CTindex 21.04). Both fiber types drove massive improvements, with 0.15% dosages yielding the highest tolerance.*




* Higher is better
Cracking
CTIndex
55
21
42
30
43
53
66
99
66
53
121
53
75
107
Control
A-0.05%
A-0.10%
A-0.15%
B-0.05%
B-0.10%
B-0.15%
3% RAS (CTIndex)
5% RAS (CTIndex)

ANOVA Results - Cracking

RAS %	Fiber Type	Fiber Dosage	p-value	Significant ($p < 0.05$) vs Control
3%	Type B	0.10%	0.0430	✓ Yes
5%	Type A	0.15%	0.0500	✓ Yes
5%	Type B	0.05%	0.0042	✓ Yes
5%	Type B	0.10%	0.0400	✓ Yes
5%	Type B	0.15%	0.0247	✓ Yes

* Only statistically significant cracking results shown. Fiber B proved universally significant at 5% RAS regardless of dosage.

IDEAL-CT Cracking Summary

- RAS Effect: Increasing RAS content from 3% to 5% severely reduced baseline cracking tolerance, with the control CTindex dropping from 55 to 21, confirming significant binder embrittlement.
- Fiber Efficacy: Both fiber types restored cracking resistance in RAS-modified mixtures, with 0.15% dosages yielding the largest CTindex improvements, especially at 5% RAS.
- Type B Superiority: Fiber B (liquid emulsion treated) consistently outperformed Fiber A in cracking resistance, achieving a peak CTindex of 119.8 at 0.10% dosage in 3% RAS mixtures.
- Statistical Significance: ANOVA confirmed significant cracking improvements for Fiber B across all dosages at 5% RAS, and at 0.10% dosage at 3% RAS, validating fiber reinforcement as a reliable cracking mitigation strategy.

Conclusions

Synthesizing findings for high-recycled, fiber-reinforced mix designs.

Conclusion

- **RAS Tradeoff Confirmed: Increasing RAS from 3% to 5% improved rut resistance but substantially reduced cracking tolerance due to binder stiffness.**
- **Fiber Efficacy: Polymer fibers restored and enhanced cracking resistance without compromising rutting stability.**
- **Optimal Performance: Fiber B (liquid emulsion treated) generally outperformed Fiber A, showing significant cracking improvements across all dosages at 5% RAS.**
- **Implementation: 25% RAP and up to 5% RAS mixtures can achieve fully balanced performance when engineered with polymer fibers.**

Recommendations for Future Work

Field Validation

Field-scale pavement sections to validate lab HWT & IDEAL-CT results under real traffic & climate conditions

Plant-Scale Production

Standardize fiber dispersion metrics; evaluate mix uniformity at production scale for quality control

Aging & Thermal Cracking

Accelerated aging (PAV + AASHTO R30) and low-temperature cracking tests on optimized fiber-RAS mixes

Higher RAS Dosages

Extend testing beyond 5% RAS with optimized fiber formulations to expand permissible use in more states

Comprehensive LCCA

Full life-cycle cost analysis: fiber premium + RAS/RAP savings + pavement life extension + EPD carbon data

Hybrid Fiber Systems

Evaluate polyester, polypropylene, and hybrid fiber blends at identified optimal RAS levels vs. aramid

Questions?

Thank you to Dr. Saadeh, Dr. Rahim, and Dr. Tanvir.

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